Kaixin Zheng

k59zheng@uwaterloo.ca | kaixin-zheng.github.io

EDUCATION

University of Waterloo — *Waterloo, ON, Canada*

Doctor of Philosophy in Computer Science — Sept. 2025–June 2029 (expected)

- Supervisor: Dr. Anita Layton.
- GPA: 91/100 (equivalent to 4.0/4.0).

Master of Mathematics in Applied Mathematics (Thesis-based) — Jan. 2023–Jan. 2025

- Supervisor: Dr. Anita Layton.
- GPA: 92.5/100 (equivalent to 4.0/4.0).
- Thesis: "Mathematical Models of Kidney Function: Effects of Hypertension and Circadian Rhythm."

New York University — *New York, NY, United States*

Bachelor of Science in Honors Mathematics; Minor in Computer Science — Sept. 2019–May 2022

- GPA: 3.89/4.0 (summa cum laude, top 5% across all majors).

University of California, Irvine — *Irvine, CA, United States*

Bachelor of Science in Physics; transferred to New York University — Sept. 2018–June 2019

- GPA: 4.0/4.0.

HONORS AND AWARDS

Women in Mathematics Mentorship Award, University of Waterloo — *Apr. 2025*

Graduate Research Studentship, University of Waterloo — *Sept. 2025–June 2029*

- Departmental research funding covering tuition and stipend for PhD studies.

International Doctoral Student Award, University of Waterloo — *Sept. 2025–June 2029*

- Multi-year award (~CAD 15,420/year) for international PhD students.

Graduate Research Studentship, University of Waterloo — *Jan. 2023–Jan. 2025*

- Departmental research funding covering full tuition and partial stipend for research-based Master of Mathematics studies.

Dean's List, New York University — *Fall 2019–Spring 2022*

Dean's Honor List, University of California, Irvine — *Fall 2018–Spring 2019*

RESEARCH EXPERIENCE

Graduate Researcher — *Jan. 2025–present*

David R. Cheriton School of Computer Science, University of Waterloo — Waterloo, ON, Canada

Supervisor: Dr. Anita Layton

- Developing scientific machine learning methods for surrogate modeling of dynamical systems.
- Investigating inverse problems, with a focus on parameter estimation in scientific dynamical systems.
- Deriving mathematical guarantees for proposed methods and analyzing failure modes of existing approaches.

Graduate Researcher — *Jan. 2023–Jan. 2025*

Applied Mathematics Department, University of Waterloo — Waterloo, ON, Canada

Supervisor: Dr. Anita Layton

- Applied numerical methods to model mammalian kidney function.
- Studied sex- and circadian-dependent effects on drug efficacy in hypertension.
- Developed nephron and renal circadian models to predict solute/water transport and analyze drug-timing effects.

Undergraduate Research Assistant *June 2021–Dec. 2021*
Courant Institute of Mathematics, New York University New York, NY, United States
Advisor: Dr. David McLaughlin

- Reviewed literature on calcium diffusion and PDE-based modeling in dendritic spines.
- Studied basic dendritic spine models and analyzed calcium dynamics.

PUBLICATIONS

Peer-reviewed

- Zheng, K., & Layton, A. T. (2026). Circadian rhythms in glomerular filtration govern natriuresis and diuretic responsiveness. *Frontiers in Physiology*, 17, 1828410. https://doi.org/10.3389/fphys.2026.1828410
- Zheng, K., & Layton, A. T. (2024). Predicting sex differences in the effects of diuretics in renal epithelial transport during angiotensin II-induced hypertension. *American Journal of Physiology–Renal Physiology*, 326(5), F737–F750. https://doi.org/10.1152/ajprenal.00398.2023

Book Review

- Abo, S., Cheung, C. C., Dasgupta, R., Dutta, P., Hakimi, S., Kaur, A., ... & Zheng, K. (2023). Featured Review: Mathematical Modeling for Epidemiology and Ecology.

COURSE PROJECTS

Project reports are available on my https://kaixin-zheng.github.io/projects/.

Polynomial Auditing for Black-Box DP Mechanisms *Winter 2026*
CS 860: Algorithms for Private Data Analysis University of Waterloo

- Proposed a polynomial upper-envelope framework for black-box auditing of differential privacy.
- Converted fitted trade-off curves into audited (ε, δ) guarantees via tangent geometry.
- Evaluated the method on Gaussian mechanisms and toy DP-SGD.

Modeling Drug Resistance in Bladder Cancer *Winter 2024*
AMATH 881: Introduction to Mathematical Oncology University of Waterloo

- Modeled spontaneous, drug-induced, and CSCs-related drug resistance in bladder cancer.
- Compared dosage strategies under the same cumulative dosage.

Lab Book on Asymptotic Analysis and Perturbation Theory *Fall 2023*
AMATH 732: Asymptotic Analysis and Perturbation Theory University of Waterloo

- Created a lab book on asymptotic expansions, perturbation methods, and signal processing.
- Provided proofs and MATLAB visualizations of normal modes in N-coupled oscillators.

TEACHING

Graduate Teaching Assistant *Sept. 2025–present*
David R. Cheriton School of Computer Science, University of Waterloo Waterloo, ON, Canada
Courses:
CS 371 Intro to Computational Mathematics *(2 terms)*

Graduate Teaching Assistant *Jan. 2023–Dec. 2024*
Applied Mathematics Department, University of Waterloo Waterloo, ON, Canada
Courses:
SYDE 211 Calculus 3
MATH 137 Honors Calculus 1
MATH 237 Advanced Calculus 3
AMATH 331 Applied Real Analysis
AMATH 382 Computational Modelling of Cellular Systems

Grader *Sept. 2021–Dec. 2021*
Courant Institute of Mathematical Sciences, New York University New York, NY, United States
Courses:
MATH-UA.251 Intro to Mathematical Modeling
MA-UY.4414 Applied Partial Differential Equations

MENTORSHIP

Undergraduate Research Mentor *Jan. 2025–Apr. 2025*
Faculty of Mathematics, University of Waterloo Waterloo, ON, Canada

VOLUNTEER

Student Representative at Math Grad Visit Day *Mar. 2024*
Applied Mathematics Department, University of Waterloo Waterloo, ON, Canada